



Management Performance: Hea 17 Acoustic conditions



Credits

No Minimum Standard

Aim

To ensure the asset provides a good indoor acoustic environment to provide comfortable conditions for asset users.

Question

Has the acoustic environment in occupied space been evaluated in terms of sound insulation, indoor ambient noise level and room acoustics?

Credits	Answer	Select all answer options that apply (C or F; D or G; E or H)
0	A.	Question not answered
0	B.	No
1	C.	Sound insulation performance standards are achieved in 50% of occupied space
1	D.	Indoor ambient noise level performance standards are achieved in 50% of occupied space
1	E.	Room acoustics performance standards are achieved in 50% of occupied space
2	F.	Sound insulation performance standards are achieved in 80% of occupied space
2	G.	Indoor ambient noise level performance standards are achieved in 80% of occupied space
2	H.	Room acoustics performance standards are achieved in 80% of occupied space

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Filtering Where an asset does not have areas used for speech or music, the room acoustics credits can be filtered out of the assessment. Rooms or areas used for speech include meeting rooms and rooms for public speaking, or rooms used for music performance and rehearsal.	E, H

Criterion	Assessment criteria	Applicable Answer
2.	Credits are awarded based on the percentage of occupied space (weighted by floor area) that meet the relevant performance standards for each space (see Methodology).	C – H
3.	A suitably qualified acoustician (SQA) has evaluated the acoustic environment in occupied space against relevant performance standards for: a) Sound insulation. b) Indoor ambient noise levels. c) Room acoustics. The requirements for a good acoustic environment will depend on the asset type and the function and uses of the different spaces within the asset. Consequently, the SQA must select appropriate asset specific performance targets to assess the acoustic environment based on current local standards, building regulations or industry best practice. Alternatively, the SQA may use the performance standards stipulated in criteria 4, 5 and 6, e.g. in the case where it can be justified that there are no appropriate local standards, building regulations or industry best practice.	C – H
4.	Indoor ambient noise levels comply with the performance standards in Table 29.	D, G
5.	The sound insulation between acoustically sensitive space and other occupied space complies with the privacy index: • $D_w + L_{AeqT} > 75$. Where privacy is very important, e.g. in a doctor's consulting room or a consulting room within a bank, or where the room is adjacent to a noisy space such as a music room, the space should comply with the enhanced privacy index: • $D_w + L_{AeqT} > 85$. D_w is the weighted sound level difference between the two spaces, and L_{AeqT} is the measured indoor ambient noise level in the acoustically sensitive room. Bedrooms in hotels must be considered acoustically sensitive rooms. For partitions and floors between bedrooms and between bedrooms and corridors, airborne sound insulation $D_{nT,w} > 50\text{dB}$.	C, F
6.	Rooms or areas used for speech (including meeting rooms and rooms for public speaking) or rooms used for music performance and rehearsal, achieve the reverberation time performance standards in Table 30 and Table 31.	E, H
7.	Performance testing should be conducted in accordance with local standards or industry best practice selected by the SQA or in accordance with the Methodology section below.	C – H

Methodology

Testing, measurement and calculation procedures

The following procedures can be followed by the SQA when measuring or calculating the levels required to demonstrate compliance with this assessment issue. Where it is necessary to deviate from the procedures, the SQA should provide a reason for doing so and confirm that the alternative procedures are adequate for demonstrating that the asset meets the acoustic performance requirements.

Measurements of airborne and impact sound insulation should be made in accordance with the relevant part of the ISO 16283 series. Measurements should be conducted between one in four pairs of adjacent rooms of each room type or performance requirements category and construction type.

Measurements of reverberation time should be made in accordance with the engineering grade (or better) requirements of ISO 3382-2 and conducted within one in four rooms of each room type of performance requirement category and construction type. In the case of room acoustics where performance standards or targets relate to sound absorption, compliance with the relevant targets can be demonstrated by confirmation from the SQA that the required amount of sound absorptive material has been applied to relevant spaces, with reference to published product or generic data.

For measurements of ambient noise, the following guidance should be used:

1. Noise from both internal sources (e.g. mechanical ventilation systems, plant noise, noise-making systems) and external sources (e.g. traffic noise transmitted via the building façade) should be included. Also, where windows are openable as part of the ventilation strategy, these should be assumed to be open for the purposes of calculations and measurements. If openable windows are present but do not form part of the background or permanent ventilation strategy, then these should be assumed to be closed for the purposes of calculations and measurements.
2. Noise from occupants and office equipment (e.g. computers) should not be included in the measurements.
3. A rate of testing of at least one in 10 rooms or spaces of each performance level shall be subject to on-site performance testing.
4. Measurements should be made in at least four rooms in which noise levels can be expected to be greatest either because they are on the noisiest façade or because they are on a naturally ventilated façade.
5. Where different ventilation strategies are used, measurements should be conducted in rooms utilising each strategy. Otherwise, measurements should be made in rooms on the noisiest façade.
6. T in $L_{Aeq,T}$ is taken as the duration of the normal working day (typically 8 hours between 0900 and 1700).
7. Measurements need not be made over a period of 8 hours if a shorter measurement period would be suitably representative. In this case, measurements should be made when external noise levels are representative of normal conditions throughout the day.
8. Measurement periods less than 30 minutes may give representative values for indoor ambient noise levels and may be utilised where this is the case. However, measurement periods shorter than 5 minutes should not be used.
9. Measurements should be taken in a minimum of three locations in rooms at a height of 1.2m above the floor level and at least 1m away from any surface.
10. The measured level of ambient noise should be used to determine compliance with the criteria for acoustically sensitive space.

Where on-site performance testing was undertaken at the time of construction or refurbishment of the asset, this information can be used to demonstrate compliance with the criteria if no changes to the asset fabric or building services have been made since the original testing, and the SQA confirms that the testing and results were in accordance with criteria 3 - 7. Where changes to the asset fabric or building services have been made since the original asset was commissioned or previous testing was performed, then the testing information needs to be supplemented to take account of the changes. The additional testing should be performed in

accordance with the guidance above and be targeted to take account of the specific alterations, e.g. sound insulation testing of replaced partitions or internal ambient noise level measurements where work has been performed on ventilation systems or the external fabric.

Calculating the percentage of occupied space that achieves the performance standards

This is based on floor area as follows:

Percentage of compliant occupied space =

$$\left(\frac{\text{Net internal area of all occupied space that achieves the relevant performance standards}}{\text{Total net internal area of all occupied space}} \right) \times 100$$

For example, for an asset with a total net internal area of all occupied space of 1,000m², at least 500m² of net internal floor area of the occupied space must meet the performance standards to achieve any credits.

Checklists and Tables

Table 29: Performance standards for indoor ambient noise levels in selected spaces

Function of space	Indoor ambient noise level (dB L_{AeqT}) *
General spaces (staffrooms, restrooms)	≤40
Single occupancy offices	≤ 0
Multiple occupancy offices	40 - 50
Meeting rooms	35 - 40
Reception areas	40 - 50
Spaces designed for speech, e.g. teaching, seminar or lecture rooms	≤35
Concert hall, theatre or auditoria	≤30
Informal café or canteen areas	≤50
Catering kitchens	≤50
Restaurant areas	40 - 55
Bars	40 - 45
Retail areas	50 - 55
Manual workshops	≤55
Sound recording studios	≤ 0
Laboratories	≤40
Sports halls or swimming pools	≤55
Library areas	40 - 50
Hotel bedrooms	<35
* Measurements must be carried out when the space is unoccupied. Where ranges of noise levels are specified and privacy is not deemed by the final occupier to be an issue, it is acceptable to disregard the lower limit of the range and consider the noise level criteria to be lower than or equal to the upper limit of the range.	

Table 30: Performance standards for reverberation time (at 500 Hz) in rooms used for speech and music

Room volume (m ³)	Reverberation time, T (seconds) *	
	Rooms for speech	Rooms for music
50	0.4	1.0
100	0.5	1.1
200	0.6	1.2
500	0.7	1.3
1000	0.9	1.5
2000	1.0	1.6

* Measurements must be carried out when the space is unoccupied. Where the above reverberation times are not appropriate for the type of space or asset being assessed, the SQA must confirm why this is the case and must set alternative appropriate reverberation times in order to demonstrate compliance.

Table 31: Performance standards for reverberation time in teaching and study spaces

Function of space	Mid-frequency reverberation time, T_{mf} (seconds) *
Open plan teaching areas	< 0.8
Open plan resource areas	< 1.0
Small lecture rooms (≤ 50 people)	< 0.8
Large lecture rooms (> 50 people)	< 1.0
Recording studio	0.6-1.2
Control room for recording	< 0.5
Libraries	< 1.0
Audio-visual and video conference rooms	< 0.8

* Measurements must be carried out when the space is unoccupied. T_{mf} is the arithmetic average of the reverberation times in the 500 Hz, 1 kHz and 2 kHz octave bands.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Records or report from SQA demonstrating the measured acoustic performance.
All	Details of the SQA responsible for the acoustic testing and assessment of the acoustic environment; including summary of their qualification(s).

Definitions

Acoustically sensitive space:

Any room or space that requires a level of privacy, which may include the following:

- a) Cellular offices
- b) Meeting, interview, consulting or treatment rooms
- c) Rooms for teaching and learning, e.g. classrooms and lecture theatres
- d) Rooms used for public speaking or seminars.

Occupied space:

A room or space within the asset that is likely to be occupied for 30 minutes or more by an asset user.

Room acoustics:

How sound behaves in an enclosed space in terms of the reverberation time (or degree of echo), overall noise levels and speech intelligibility. Room acoustics are influenced by room geometry and distribution of acoustic absorption either through the general room finishes or through the introduction of sound absorbing products.

Suitably qualified acoustician:

An individual achieving all the following items is considered as 'suitably qualified' for the purposes of this assessment issue:

- Has a minimum of three years' relevant experience (within the last five years) with such experience clearly demonstrating a practical understanding of factors affecting acoustics in relation to construction and the built environment; including acting in an advisory capacity to provide recommendations for suitable acoustic performance levels and mitigation measures.
- An individual who holds a recognised acoustic qualification and membership of an appropriate professional body.

Where an SQA is verifying the acoustic measurements or calculations carried out by another acoustician who does not meet the SQA requirements, they must, as a minimum, have read and reviewed the report and confirm in writing that they have found it to:

- Represent sound industry practice
- Be appropriate given the building being assessed and scope of works proposed
- Avoid invalid, biased and exaggerated recommendations.

Additionally, written confirmation from the third party verifier that they comply with the definition of an SQA is required.